

CONSTRAINT HANDLING: ALLOWING INFEASIBILITY

Lecture 32: Presented by Jacomine Grobler



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Allowing infeasibility

Constraint handling techniques allowing a (meta)heuristic search to venture into the infeasible region of the optimisation problem under consideration are easily implementable, but the challenge lies in the selection of an appropriate value for penalty weight.

If this value is too large, the search may never enter the infeasible region (in the case of a stochastic (meta)heuristic search), while if it is too small the search may never return a feasible solution upon termination.

The (population of) initial solution(s) does not itself have to be feasible when applying this type of constraint handling technique, but incumbent updates should, of course, only occur for feasible solutions.

Example 7.6 (pp. 227–231)

Consider the non-linear optimisation problem of

$$\text{maximising } f_{7.3}(x_1, x_2) = \exp [-(x_1 - 3)^2 - (x_2 - 2)^2]$$

subject to the constraints

$$\begin{aligned}x_1 x_2 &= 5, \\x_1, x_2 &\geq 0.\end{aligned}$$

Approximate the solution to the constrained optimisation problem by applying a local search heuristic and using a quadratic penalty function. We take $\rho = 10$, the grid lattice discretisation cell lengths as $\varepsilon = 0.25$ initially and start at the initial (infeasible) solution $\mathbf{x}^0 = [0.5 \ 0.1]^T$. We work accurately to three decimal places and halve the lattice cell lengths each time the search stagnates at a local optimum, until these lengths drop below $\varepsilon_{\text{tol}} = 0.001$ for the first time.